

CHE 311

1. Course number and name

CHE 311. Transport Phenomena I

2. Credits, contact hours and categorization

Credit hours: 3; Contact Hours: 150 min/week lecture; Engineering

3. Instructor's or course coordinator's name

Ying Liu, Vivek Sharma, Gang Cheng

4. Text book, title, author, publisher, year

R.B.Bird, W.E. Stewart & E. N. Lightfoot, Transport Phenomena, 2nd Ed., Wiley (2006)

5. Specific course information

a. Brief description of the content of the course (catalog description)

Momentum transport phenomena in chemical engineering. Fluid statics. Fluid mechanics; laminar and turbulent flow; boundary layers; flow over immersed bodies.

b. Prerequisites or co-requisites

Credit or concurrent registration in CHE 205, CHE 210, and MATH 220.

c. Indicate whether a required, elective, or selected elective.

Required.

6. Specific goals for the course

a. Performance indicators of student learning.

- Calculate scalar, vector, and tensor operations.
- Understand the essential physical principles of momentum transport and definition of viscosity.
- Set up differential momentum balance and analyze boundary conditions.
- Solve constant-coefficient ODEs and apply boundary conditions to determine steady state one-dimensional flow velocity and stress profiles.
- Solve PDEs by using the methods of separation of variables and combination of variables to result the velocity profiles of the flow field which has two independent variables.
- Formulate and solve the mathematical models to show how viscosity of a Newtonian fluid can be measured by various types of viscometers. Analyze the advantage and disadvantage of each viscometer.
- Apply time-smoothed equations of change for turbulent flow.
- Analyze the time-smoothed velocity profile near a wall.
- Understand the physical meaning and mathematical expression of the empirical expressions for the turbulent momentum flux.
- Understand the definition of friction factor. Use Moody diagram to solve various pipe-design problems.
- Set up macroscopic mass, momentum, and mechanical energy balances for isothermal flow systems.

a. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcome 1 Not Covered, 2 Introduced, 3 Reinforced, 4 Emphasized	1	2	3	4
(1) an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.			X	
(2) an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.		X		
(3) an ability to communicate effectively with a range of audiences.			X	
(4) an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.		X		
(7) an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.		X	X	

7. Brief list of topics to be covered

Week #1: Introduction and mathematical preliminaries

Scalars, vectors, and tensors—operations involving them

Week #2: Transport coefficients (Chap. 1, 9, 17)

Fourier's law of heat conduction

Fick's law of binary diffusivity

Newton's law of viscosity

Week #3: Shell Momentum balances and velocity distribution in laminar flow

Week #4: Apply shell momentum balances on more representative laminar flows

Week #5: Conservation equations

Differential equations for mass and linear momentum

Week #6: Apply differential mass and linear momentum balance on representative flow problems

Week #7: Conservation equations for angular momentum* and energy balance

Week #8: Macroscopic balances for mass and momentum

Week #9: Macroscopic balances for mechanical energy

Week #10: Velocity distributions with more than one independent variable

Time dependence

Stream function

Week #11: Similarity analysis

Week #12: Introduction of turbulent flow

Week #13: Time-smoothed velocity profile for turbulent flow near a wall

Week #14: Friction factors for flow in a circular tube

Week #15: Friction factors for flow around a sphere